

ÖMER FARUK KOÇ

MLOps & AI Platform Engineer

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SUMMARY

MLOps and AI Platform Engineer with 3+ years of experience building and operating production ML, data, and Generative AI systems in financial services. Owned production ML model lifecycle and serving workflows, engineered large-scale customer-level data platforms, and reduced a core feature-engineering workflow runtime by **75%**. Recent engineering work focuses on production-oriented Agentic AI, multilingual RAG evaluation, deterministic guardrails, observability, and failure recovery alongside ML platform engineering and on-premises LLM infrastructure.

EXPERIENCE

MLOps & Analytics Engineer

Fibabanka — Analytics Center of Excellence

Mar 2023 – Mar 2026

Istanbul, Türkiye

- Re-architected sequential Oracle feature-engineering workflows into parallel, modular **dbt + Airflow** pipelines, reducing runtime by **75%** while improving lineage, dependency visibility, and maintainability.
- Engineered batch and near-real-time pipelines consolidating multiple source systems into reusable **customer-level feature, training, inference, and analytics datasets** supporting production ML workflows at scale.
- Owned deployment and lifecycle management for a **portfolio of production ML models** on **MLRun (Iguazio)**, covering versioning, validation, production promotion, retraining, monitoring, and containerized FastAPI serving across Docker and Kubernetes environments.
- Built a **centralized feature platform** and dimensional data models shared across production ML use cases, replacing duplicated project-specific feature pipelines with reusable, production-ready features.
- Prototyped ML and LLM workloads in Azure and migrated validated workloads to **on-premises GPU infrastructure**, serving quantized open-source models with llama.cpp and Ollama while preserving internal data residency.
- Delivered **RAG, document-processing, and structured information extraction** workflows and engineered a **GPU-distributed call-center intelligence pipeline** for high-volume daily processing, combining WhisperX transcription, timestamp alignment, speaker diarization, LLM summarization, and structured KPI extraction.
- Implemented data-quality, drift, and inference monitoring with **Great Expectations and Deepchecks**; automated model/data workflows with Airflow and CI/CD with GitHub Actions.

TECHNICAL SKILLS

Programming & Backend: Python, SQL, FastAPI, REST APIs, Pydantic, SQLAlchemy

MLOps & AI Platform: MLRun, Docker, Kubernetes, GitHub Actions, Model Serving, Feature Stores, Canary Deployment, Model Monitoring, Drift Detection

Generative AI & Agentic Systems: LangGraph, LangChain, RAG, Hybrid Retrieval, Qdrant, Deterministic Guardrails, Human-in-the-Loop, Evaluation, MCP, llama.cpp, Hugging Face Transformers, WhisperX

Data & Streaming: Apache Airflow, dbt, Apache Kafka, Redis, PostgreSQL, Oracle, Trino/Starburst, ETL/ELT

Observability & Infrastructure: OpenTelemetry, Jaeger, Prometheus, Grafana, Great Expectations, Deepchecks, Azure, On-Premises GPU, Linux, Git

SELECTED PROJECTS

Agentic Customer Service Platform [GitHub]

2026

Built a production-oriented Agentic AI control plane where **LLM outputs are treated as untrusted proposals** and business mutations remain behind deterministic authorization, policy, confirmation, revalidation, and idempotency boundaries. Across **540 measured semantic/safety attempts**, reduced unsafe executable survivors to **0**; passed **18/18 operational release scenarios** and observed **0 unauthorized mutations, confirmation bypasses, or duplicate effects** in a 100-sample real-LLM adversarial audit.

Knowledge Base RAG [GitHub]

2026

Engineered a multilingual RAG platform with **hybrid dense and sparse retrieval, RRF fusion, cross-encoder reranking, evidence validation, tenant isolation, and citation controls**. On a frozen **220-question multilingual benchmark**, a multilingual reranker increased Cross MRR from **0.367 to 0.956** versus the prior English reranker and achieved **100% retrieval Recall@5** on both TR-to-EN and EN-to-TR slices; end-to-end evaluation reached **95.9% candidate evidence recall** with **0 citation-contract failures**.

ModelOps Control Plane [GitHub]

2026

Built a policy-driven ML release control plane with **weighted canary routing, delayed-ground-truth quality evaluation, automated promotion/rollback, fault injection, and desired-vs-observed reconciliation**. Validated automatic rollback under injected latency and quality regressions, self-healing router state after restart, and progressive **10% to 25% to 50% to 100%** traffic promotion across a multi-service deployment stack.

Real-Time Commerce Platform [GitHub]

2026

Designed and optimized an at-least-once Kafka processing platform with **idempotent consumers, transactional persistence, Redis coordination, DLQ handling, and a transactional outbox**. Reduced Kafka offset commit calls by **approximately 28.6x**, cut a critical PostgreSQL lookup from **10.9 ms to 0.25 ms**, and increased measured sustainable local throughput from **approximately 750 to 1,050 events/s (+40%)** while preserving **idempotency and durable processing guarantees**.

EDUCATION

B.Sc. in Electronics Engineering, Gebze Technical University